

An Analysis of Regularization machine learning

Regularization machine learning

Regularization machine learning -- Part 1

When labels are more expensive to gather than input examples, semi-supervised learning can be useful. Regularizers have been designed to guide learning algorithms to learn models that respect the structure of unsupervised training samples. If a symmetric weight matrix W is given, a regularizer can be defined:

Regularization machine learning -- Part 2

Early stopping can be viewed as regularization in time. Intuitively, a training procedure such as gradient descent tends to learn more and more complex functions with increasing iterations. By regularizing for time, model complexity can be controlled, improving generalization. Early stopping is implemented using one data set for training, one statistically independent data set for validation and another for testing. The model is trained until performance on the validation set no longer improves and then applied to the test set.

Regularization machine learning -- Part 3

In mathematics, statistics, finance, and computer science, particularly in machine learning and inverse problems, regularization is a process that converts the answer to a problem to a simpler one. It is often used in solving ill-posed problems or to prevent overfitting. Although regularization procedures can be divided in many ways, the following delineation is particularly helpful:

Regularization machine learning -- Part 4

Group sparsity with overlaps The algorithm described for group sparsity without overlaps can be applied to the case where groups do overlap, in certain situations. This will likely result in some groups with all zero elements, and other groups with some non-zero and some zero elements. If it is desired to preserve the group structure, a new regularizer can be defined:

Regularization machine learning -- Part 5

Classification Empirical learning of classifiers (from a finite data set) is always an underdetermined problem, because it attempts to infer a function of any x given only examples $x_1, x_2, \dots, x_n$. A regularization term (or regularizer) $R(f)$ is added to a loss function:

Regularization machine learning -- Part 6

Regularization can be motivated as a technique to improve the generalizability of a learned model. The goal of this learning problem is to find a function that fits or predicts the outcome (label) that minimizes the expected error over all possible inputs and labels. The expected error of a function f_n is:

Regularization machine learning -- Part 7

This regularizer constrains the functions learned for each task to be similar to the overall average of the functions across all tasks. This is useful for expressing prior information that each task is expected to share with each other task. An example is predicting blood iron levels measured at different times of the day, where each task represents an individual.

Regularization machine learning -- Part 8

The exact solution to the unregularized least squares learning problem minimizes the empirical error, but may fail. By limiting T , the only free parameter in the algorithm above, the problem is regularized for time, which may improve its generalization. The algorithm above is equivalent to restricting the number of gradient descent iterations for the empirical risk

Regularization machine learning -- Part 9

These techniques are named for Andrey Nikolayevich Tikhonov, who applied regularization to integral equations and made important contributions in many other areas. When learning a linear function f , characterized by an unknown vector w such that $f(x) = w \cdot x$, one can add the L_2 -norm of the vector w to the loss expression in order to prefer solutions with smaller norms. Tikhonov regularization is one of the most common forms. It is also known as ridge regression. It is expressed as:

Regularization machine learning -- Part 10

$$R(f_1 \dots f_T) = \sum_{t=1}^T \|f_t - \bar{f}\|^2 + \sum_{s=1}^T f_s^T H_k \sum_{t=1}^T f_t$$

Regularization machine learning -- Part 11

Regularization in machine learning In machine learning, a key challenge is enabling models to accurately predict outcomes on unseen data, not just on familiar training data. Regularization is crucial for addressing overfitting—where a model memorizes training data details but cannot generalize to new data. The goal

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of regularization is to encourage models to learn the broader patterns within the data rather than memorizing it. Techniques like early stopping, L1 and L2 regularization, and dropout are designed to prevent overfitting and underfitting, thereby enhancing the model's ability to adapt to and perform well with new data, thus improving model generalization.

methods of regularization to the linear model are:

Regularization machine learning -- Part 12

$$\begin{aligned} w_T &= (I - \gamma_n X^T X)^{-1} \gamma_n \sum_{i=0}^{T-1} (I - \gamma_n X^T X)^i X^T Y + \gamma_n X^T Y = \gamma_n \sum_{i=0}^{T-1} (I - \gamma_n X^T X)^i X^T Y + \gamma_n X^T Y \\ &= \gamma_n \sum_{i=0}^{T-1} (I - \gamma_n X^T X)^i X^T Y + \gamma_n X^T Y \end{aligned} \quad \{\displaystyle \begin{aligned} w_T &= \left(I - \frac{\gamma_n}{n} X^T X \right)^{-1} \frac{\gamma_n}{n} \sum_{i=0}^{T-1} \left(I - \frac{\gamma_n}{n} X^T X \right)^i X^T Y + \frac{\gamma_n}{n} X^T Y \\ &= \frac{\gamma_n}{n} \sum_{i=0}^{T-1} \left(I - \frac{\gamma_n}{n} X^T X \right)^i X^T Y + \frac{\gamma_n}{n} X^T Y \end{aligned} \}$$

Regularization machine learning -- Part 13

L1 regularization (also called LASSO) leads to sparse models by adding a penalty based on the absolute value of coefficients. L2 regularization (also called ridge regression) encourages smaller, more evenly distributed weights by adding a penalty based on the square of the coefficients.

Regularization machine learning -- Part 14

$$\min_{\mathbf{f}} \sum_{i=1}^n V(\mathbf{f}(\mathbf{x}^i), y^i) + \lambda \|\mathbf{f}\|_H^2 \quad \{\displaystyle \min_{\mathbf{f}} \sum_{i=1}^n V(\mathbf{f}(\mathbf{x}_i), y_i) + \lambda \|\mathbf{f}\|_H^2 \}$$

Regularization machine learning -- Part 15

Other uses of regularization in statistics and machine learning Bayesian learning methods make use of a prior probability that (usually) gives lower probability to more complex models. Well-known model selection techniques include the Akaike information criterion (AIC), minimum description length (MDL), and the Bayesian information criterion (BIC). Alternative methods of controlling overfitting not involving regularization include cross-validation. Examples of applications of different